AUTOMATED STATISTICAL ANALYSIS PIPELINE

statistical_pipeline.R

Comprehensive Multi-Group Analysis · Version 1.0.0

Purpose	Automate the full statistical workflow: descriptives → normality → group tests → effect sizes → power → correlation → regression → export
Input	Any R data.frame (CSV, Excel, RDS, or built-in example)
Output	Excel workbook · CSV tables · PNG plots · Word report · RDS cache
Language	R ≥ 4.2 (tested on 4.3, 4.4)

1. Overview

The pipeline is a single self-contained R script that, given a data frame and a small configuration block, performs every common statistical analysis task needed for group-comparison studies. It is designed to work out-of-the-box with zero manual steps: set the CONFIG, run source(), receive a fully organised output folder.

Design principles:

- Zero-config default — runs on a synthetic example immediately after sourcing.
- Resilient — every analysis block is wrapped in tryCatch; one failure never stops the rest.
- Auditable — every decision (parametric vs non-parametric, which post-hoc to flag as primary) is logged and stored.
- Reproducible — set.seed is used where needed; full RESULTS list is saved as RDS.
- Modular — individual sections can be toggled with run_* flags in CONFIG.

2. Quick Start

Step 1 — Open statistical_pipeline.R in RStudio or any R editor.

Step 2 — Edit the CONFIG list (Section 1 of the script). At minimum set data_source and group_var.

Step 3 — Run:

```
source("statistical_pipeline.R")
```

Step 4 — Inspect outputs in the folder specified by output_dir (default: pipeline_output/ in the working directory).

Built-in example (no data needed):

```
# No changes required — just source the file as-is
source("statistical_pipeline.R")
# Generates 90-row, 3-group synthetic dataset and runs everything
```

3. Configuration Reference

All user-facing settings live in the CONFIG list at the top of the script. Every parameter has a sensible default.

3.1 Data Source

Parameter	Default	Type	Description
data_source	"builtin_example"	character	"builtin_example" | "csv" | "excel" | "rds"
data_path	""	character	Path to file when data_source is not builtin_example

3.2 Variable Roles

Parameter	Default	Type	Description
group_var	"group"	character	Column name of the categorical grouping variable
outcome_vars	NULL	character[]	NULL = auto-detect all numeric columns (excluding special vars)
covariate_vars	NULL	character[]	Numeric covariates for ANCOVA; NULL = skip ANCOVA
id_var	NULL	character	Subject ID for repeated-measures models
time_var	NULL	character	Time column for survival or longitudinal analysis
event_var	NULL	character	0/1 event indicator for survival analysis

3.3 Analysis Toggles

Parameter	Default	Type	Description
run_descriptives	TRUE	logical	Per-group mean, SD, SE, median, IQR, CI, skewness, kurtosis
run_normality	TRUE	logical	Shapiro-Wilk, KS-Lilliefors, Anderson-Darling, Jarque-Bera
run_group_tests	TRUE	logical	t-test / Welch / Mann-Whitney or ANOVA / Welch-ANOVA / Kruskal-Wallis

run_posthoc	TRUE	logical	Tukey, Games-Howell, Dunn, pairwise Wilcoxon with p-adjustment
run_effect_sizes	TRUE	logical	Cohen's d, Hedges' g, Glass's Δ, rank-biserial r, η^2, ω^2, ε^2
run_power	TRUE	logical	Observed power + n needed for 80% and 90% power
run_correlation	TRUE	logical	Pearson and Spearman with significance; corrplot + scatter matrix
run_regression	TRUE	logical	Simple and full OLS regression per outcome with diagnostics
run_ancova	FALSE	logical	ANCOVA with emmeans post-hoc; requires covariate_vars
run_nonparametric	TRUE	logical	Non-parametric branch is always available alongside parametric
run_survival	FALSE	logical	KM curves, log-rank, Cox PH; requires time_var + event_var
run_repeated	FALSE	logical	LME4 mixed model; requires id_var + time_var
run_missing_analysis	TRUE	logical	Per-variable missing count, %, pattern plot

3.4 Thresholds

Parameter	Default	Type	Description
alpha	0.05	numeric	Significance threshold for all tests
normality_alpha	0.05	numeric	Threshold for normality decision (often set higher, e.g. 0.10)
p_adjust_method	"BH"	character	"bonferroni" | "holm" | "BH" | "fdr" | "none"
min_n_per_group	3	integer	Groups below this n are skipped
outlier_threshold	3	numeric	Flag observations where |z-score| exceeds this value

3.5 Visualization & Export

Parameter	Default	Type	Description
color_palette	"Set2"	character	Any RColorBrewer palette name
plot_width	10	numeric	Plot width in inches
plot_height	7	numeric	Plot height in inches

plot_dpi	300	integer	Resolution for saved PNG files
theme_choice	"bw"	character	"bw" | "minimal" | "classic" | "dark"
output_dir	"pipeline_output"	character	Root folder for all outputs
export_excel	TRUE	logical	Write multi-sheet Excel workbook
export_csv	TRUE	logical	Write individual CSV files per result table
export_plots	TRUE	logical	Save all plots as PNG
export_report	TRUE	logical	Generate Word .docx summary report
verbose	TRUE	logical	Print timestamped progress messages to console

4. Pipeline Stages

4.1 Data Loading (Section 3)

Reads data from the source specified in CONFIG, then passes it through janitor::clean_names() to sanitise all column names (spaces become underscores, etc.). Group variable is coerced to factor. Outcome variables are auto-detected as all numeric columns not assigned a special role.

4.2 Missing Data Analysis (Section 4)

Uses naniar to compute per-variable counts and percentages of missing values. A visual bar chart is saved. The pipeline does not auto-impute — it flags and informs. To impute before re-running add:

```
df <- complete(mice(df, m=5, method="pmm"), 1)
```

4.3 Descriptive Statistics (Section 5)

Computes the following for each outcome variable, broken down by group:

Statistic	Description
n, n_missing	Sample size and missing count
mean, sd, se	Arithmetic mean, standard deviation, standard error
median, iqr, q25, q75	Median and interquartile range with quartiles
min, max, range	Minimum, maximum, and total spread
cv_pct	Coefficient of variation as a percentage
skewness, kurtosis	Moments of the distribution (via moments package)
ci_lower, ci_upper	95% confidence interval for the mean

Additionally, outliers are flagged for any observation where |z-score| exceeds outlier_threshold. Visualizations include violin + box + beeswarm plots per variable and a ridge-density panel for all variables scaled to the same axis.

4.4 Normality Tests (Section 6)

Four complementary normality tests are run on each group × outcome combination:

Test	When used / Notes
Shapiro-Wilk (SW)	n = 3–5000; most powerful for small samples; primary decision driver
Kolmogorov-Smirnov / Lilliefors (KS)	n ≥ 4; corrected for estimated parameters
Anderson-Darling (AD)	n ≥ 7; weights tails more heavily than KS
Jarque-Bera (JB)	n ≥ 8; tests skewness and kurtosis jointly

The pipeline decides normality_assumed = TRUE if the majority of groups pass Shapiro-Wilk at normality_alpha. QQ-plots (sample vs theoretical quantiles) are saved for each outcome.

4.5 Group Comparison Tests (Section 7)

The pipeline branches automatically based on number of groups and normality assumption:

Scenario	Tests run
2 groups, normal, equal var	Student t-test (primary: Welch t-test)
2 groups, normal, unequal var	Welch t-test
2 groups, non-normal	Mann-Whitney U (Wilcoxon rank-sum)
3+ groups, normal, equal var	One-Way ANOVA
3+ groups, normal, unequal var	Welch One-Way ANOVA
3+ groups, non-normal	Kruskal-Wallis H

Levene's test and Bartlett's test are both run for variance homogeneity. Both parametric and non-parametric results are always computed and stored regardless of which branch is chosen as primary. P-values are adjusted across outcomes using the method in p_adjust_method.

4.6 Post-hoc Tests (Section 8)

Applied when there are 3 or more groups. Four procedures are run simultaneously:

Method	Use case
Tukey HSD	Parametric, equal variances — classic pairwise with FWER control
Games-Howell	Parametric, unequal variances — no assumption of homoscedasticity
Dunn test (BH adjusted)	Non-parametric, follows Kruskal-Wallis; primary for skewed data
Pairwise Wilcoxon	Non-parametric, flexible; p-adjustment via p_adjust_method

The recommended field in RESULTS$posthoc[[var]] points to the best-fit method based on the normality and variance decisions made in earlier stages. Significance bracket plots are generated for each outcome.

4.7 Effect Sizes (Section 9)

Measure	Interpretation scale
Cohen's d (2 groups)	< 0.2 negligible · 0.2–0.5 small · 0.5–0.8 medium · > 0.8 large
Hedges' g (2 groups)	Bias-corrected version of Cohen's d; preferred for unequal n
Glass's Δ (2 groups)	Uses control group SD; preferred when SDs differ substantially
Rank-biserial r (2 groups)	Non-parametric ES for Mann-Whitney; range −1 to +1
Eta-squared η^2 (3+ groups)	< 0.01 negligible · 0.01–0.06 small · 0.06–0.14 medium · > 0.14 large
Omega-squared ω^2 (3+ groups)	Less biased than η^2; recommended for small samples
Epsilon-squared ε^2 (3+ groups)	Another less-biased η^2 alternative
KW epsilon-squared (3+ groups)	Non-parametric effect size for Kruskal-Wallis

4.8 Power Analysis (Section 10)

Uses the pwr package to compute:

- observed_power — the probability of detecting the observed effect at current n
- n_for_80pct_power — minimum n per group needed for 80% power
- n_for_90pct_power — minimum n per group needed for 90% power

Effect sizes are taken from the effect size stage (Cohen's d for two groups, Cohen's f derived from η^2 for ANOVA). If effect sizes are unavailable, conventional medium values (d = 0.5, f = 0.25) are substituted.

4.9 Correlation Analysis (Section 11)

Both Pearson (linear, parametric) and Spearman (rank-based, non-parametric) correlations are computed with associated p-values using Hmisc::rcorr. Three visualizations are produced:

- Pearson corrplot — ellipses (upper triangle) + numeric coefficients (lower triangle), non-significant cells marked
- Spearman corrplot — same layout for rank correlations
- GGally scatter matrix — pairwise scatter plots with group coloring, density diagonals, and correlation coefficients

4.10 Regression Analysis (Section 12)

For each outcome variable, two OLS regression models are fitted:

Model	Formula
Simple model	outcome ~ group
Full model	outcome ~ group + all_other_outcomes

Both models produce broom::tidy() coefficient tables (estimate, SE, t, p, 95% CI) and broom::glance() fit statistics (R^2, adj-R^2, F, df, AIC, BIC). Standard diagnostic plots (residuals vs fitted, QQ, scale-location, leverage) are saved for the full model.

4.11 ANCOVA (Section 13, optional)

Activated by setting run_ancova = TRUE and providing covariate_vars. Fits the model outcome ~ group + covariate(s) via aov(), then uses emmeans to compute group means adjusted for covariates and pairwise comparisons. Effect sizes (η^2) are computed per term.

4.12 Repeated Measures / Mixed Models (Section 14, optional)

Activated by setting run_repeated = TRUE and providing id_var and time_var. Fits a linear mixed-effects model using lmerTest::lmer with random intercepts by subject and fixed effects of time × group. Reports fixed-effect ANOVA table, ICC, and both marginal (R^2m) and conditional (R^2c) R-squared values.

4.13 Survival Analysis (Section 15, optional)

Activated by setting run_survival = TRUE and providing time_var and event_var. Produces:

- Kaplan-Meier curves with confidence intervals and at-risk table
- Log-rank test (survdiff) for group comparison
- Cox proportional hazards model with hazard ratios (exp(β)) and 95% CI

5. Output Structure

All outputs land in the folder specified by output_dir. The structure is:

```
pipeline_output/
  ├── RESULTS.rds                  # Complete R object — all results
  ├── statistical_report.docx      # Word summary report
  ├── plots/
  │   ├── dist_{var}.png           # Violin + box + beeswarm per variable
  │   ├── qq_{var}.png             # QQ plots per variable
  │   ├── posthoc_{var}.png        # Post-hoc significance brackets
  │   ├── ridge_distributions.png  # All outcomes ridge plot
  │   ├── mean_se_barplot.png      # Grouped bar chart
  │   ├── pvalue_heatmap.png       # Heatmap of raw vs adjusted p
  │   ├── correlation_pearson.png  # Pearson corrplot
  │   ├── correlation_spearman.png # Spearman corrplot
  │   ├── scatter_matrix.png       # Pairwise scatter (GGally)
  │   ├── regression_diagnostics_{var}.png
  │   ├── missing_values.png
  │   └── survival_km.png          # (if survival enabled)
  └── tables/
      ├── full_results.xlsx        # All tables in one workbook
      ├── descriptives.csv
      ├── normality.csv
      ├── group_tests.csv
      ├── effect_sizes.csv
      ├── power.csv
      ├── outliers.csv
      └── missing.csv
```

6. RESULTS Object Reference

The RESULTS list is returned invisibly from source() and saved as pipeline_output/RESULTS.rds. Reload any time with:

```
RESULTS <- readRDS("pipeline_output/RESULTS.rds")
```

RESULTS field	Contents

$data	Cleaned data frame (factor group, clean column names)
$skim	skimr skim summary of all variables
$descriptives	Per-variable × group tibble with all descriptive stats
$outliers	Rows where |z-score| > outlier_threshold
$missing	naniar per-variable missing summary
$normality	4-test normality results per variable × group
$group_tests	All test statistics, raw and adjusted p-values, significance flags
$posthoc[[var]]	List with tukey / games_howell / dunn / pairwise_wilcox / recommended
$effect_sizes	All effect size estimates with interpretation labels
$power	Observed power and required n for 80% and 90% power
$correlation	List with pearson and spearman Hmisc::rcorr objects
$regression[[var]]	simple_model, simple_tidy, full_model, full_tidy, glance objects
$ancova[[var]]	(optional) ANCOVA model, emmeans, eta²
$repeated_measures[[var]]	(optional) lmerTest model, ANOVA table, ICC, R²
$survival	(optional) km_fit, logrank, cox_fit, cox_tidy

7. Automatic Statistical Decision Logic

The pipeline makes test selection transparent. Every decision is documented in RESULTS$group_tests (columns: normality_assumed, var_homogeneous, test_used, primary_pval).

NOTE	The pipeline always runs both parametric and non-parametric tests in parallel. 'Primary' simply means the test most appropriate given the data properties. All others are stored and available.

Decision flowchart (simplified):

- Normality assumed if majority of Shapiro-Wilk tests pass at normality_alpha.
- Variance homogeneity assumed if Levene's test p > alpha.
- 2 groups, normal → Welch t-test (robust to slight variance inequality).
- 2 groups, non-normal → Mann-Whitney U.
- 3+ groups, normal, homoscedastic → One-Way ANOVA + Tukey HSD.
- 3+ groups, normal, heteroscedastic → Welch ANOVA + Games-Howell.
- 3+ groups, non-normal → Kruskal-Wallis + Dunn test.

8. Package Dependencies

All required packages are installed automatically on first run. A full internet connection is needed only the first time. Subsequent runs use cached installations.

Category	Packages
Core data manipulation	tidyverse, data.table, janitor, skimr
Statistical tests	rstatix, coin, DescTools, PMCMRplus, nortest, moments

Effect sizes	effectsize
Power analysis	pwr
Mixed models	lme4, lmerTest, MuMIn
Regression	car, MASS, broom
Correlation	corrplot, Hmisc, psych
Survival	survival, survminer
Missing data	naniar, mice
Visualization	ggplot2, ggpubr, ggbeeswarm, ggridges, patchwork, viridis, RColorBrewer, GGally
Export	openxlsx, flextable, officer, knitr, kableExtra
ANCOVA / emmeans	emmeans

9. Using Your Own Data

9.1 CSV or Excel

```
CONFIG$data_source  <- "csv"
CONFIG$data_path    <- "path/to/your/data.csv"
CONFIG$group_var    <- "treatment_arm"
CONFIG$outcome_vars <- c("score", "weight", "bp_systolic")
source("statistical_pipeline.R")
```

9.2 Data frame already in R

```
# Option: write to CSV first
write.csv(my_df, "my_data.csv", row.names = FALSE)
CONFIG$data_source <- "csv"
CONFIG$data_path   <- "my_data.csv"
```

Alternatively, replace the builtin_example branch in load_data() with df <- my_df.

9.3 Tips for Best Results

- Ensure group_var has clean, consistent group labels (no leading/trailing spaces).
- For survival analysis, event_var must be strictly 0 (censored) or 1 (event).
- For repeated measures, each unique id_var × time_var combination must be a single row.
- Outcome variables should be on a continuous numeric scale. Binary or count outcomes need logistic/Poisson — consider extending the regression section.
- If normality tests all fail, increase normality_alpha to 0.10 or switch primary decisions to non-parametric by hardcoding normality_assumed = FALSE.

10. Interpretation Guide

10.1 Reading group_tests Output

Column	Meaning
primary_pval	Raw p-value from the recommended test
primary_pval_adj	P-value after multiple testing adjustment
significant	TRUE if primary_pval < alpha
significant_adj	TRUE if primary_pval_adj < alpha (preferred)
test_used	Which test was chosen as primary

10.2 Effect Size Benchmarks

Measure	Small
Cohen's d / Hedges' g	0.2
Eta-squared η^2	0.01
Omega-squared ω^2	0.01
Cohen's f (ANOVA)	0.10
Rank-biserial r	0.10

10.3 Power Interpretation

- Power < 0.60 → high risk of Type II error; increase sample size.
- Power 0.60–0.80 → borderline; interpret negative results cautiously.
- Power ≥ 0.80 → conventional acceptable threshold.
- Power ≥ 0.90 → high confidence; preferred for clinical or policy research.

11. Extending the Pipeline

The script is deliberately structured so new analyses slot in cleanly. Pattern to add a new section:

```
# Section XX: My New Analysis
if (CONFIG$run_my_analysis) {
  log_msg("=== MY NEW ANALYSIS ===")
  RESULTS$my_analysis <- safe_run('my_analysis', {
    # ... your code ...
  })
}
```

Then add run_my_analysis = FALSE to CONFIG defaults, set to TRUE to activate, and add the result to the Excel workbook and Word report export sections.

Common extensions:

- Logistic regression: use glm(outcome ~ group, family=binomial)
- Poisson/negative binomial: glm or MASS::glm.nb for count outcomes
- Bayesian tests: BayesFactor package for Bayes factor alternatives
- Machine learning: caret or tidymodels for predictive modelling
- Time series: tseries or forecast for longitudinal without random effects

12. Troubleshooting

Problem	Solution
Package install fails	Run install.packages('pkg') manually with admin rights, or use renv for project-level control
'object not found' error	Ensure CONFIG$group_var matches the actual column name exactly (case-sensitive after clean_names)
ANOVA fails with < 3 obs	Check min_n_per_group; increase if groups have very few observations
Corrplot shows all grey	All p-values > alpha; reduce alpha or check that variables actually vary
Word report not generated	Ensure officer and flextable are installed; check file permissions on output_dir
Welch ANOVA gives warning	Normal; Welch handles unequal variances but still assumes approximate normality
Survival plot fails	Check that event_var contains exactly 0 and 1 values with no NAs
LME4 model fails to converge	Try adding optimizer = lmerControl(optimizer='bobyqa') to the lmer() call

13. Reproducibility & Citation

To ensure reproducibility, the pipeline saves the complete RESULTS object as an RDS file. To include in a methods section, report:

- R version (R.version.string)
- Key package versions: sessionInfo() after sourcing the pipeline
- CONFIG settings used
- Random seed (42 for built-in example; set via set.seed() for custom data)

Example methods paragraph:

"Data were analysed using R (v4.x) with an automated statistical pipeline. Normality was assessed using Shapiro-Wilk tests. Group differences were evaluated using [test_used] with p-values adjusted using the Benjamini-Hochberg procedure. Effect sizes were estimated as [Cohen's d / η^2] and interpreted per Cohen (1988). Power analysis was conducted using the pwr package. Statistical significance was defined as adjusted $p < 0.05$."